Uncertainty-Aware AI Triage for Critical Findings in Medical Imaging

Calibrated Confidence Scoring for Radiology Worklist Prioritization with Monte Carlo Dropout and PACS-Native Integration

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. Abstract

Artificial intelligence systems for radiological image interpretation have demonstrated strong diagnostic performance in controlled settings, yet clinical adoption remains limited. A central barrier is the absence of calibrated uncertainty information: most deployed systems produce binary predictions without communicating the confidence of their outputs, leaving clinicians unable to distinguish reliable from unreliable assessments. This gap undermines trust, complicates regulatory evaluation, and creates patient safety concerns when false-negative predictions go unrecognized.

We propose a medical imaging triage system that combines state-of-the-art visual feature extraction (Vision Transformer or ConvNeXt architectures) with Monte Carlo Dropout-based epistemic uncertainty estimation to produce calibrated confidence scores for critical radiological findings, including pneumothorax, pulmonary embolism, and acute stroke. The system routes predictions through a decision framework: high-confidence critical findings trigger immediate clinician alerts, high-confidence normal findings are deprioritized, and low-confidence cases are escalated to senior radiologist review.

This concept paper presents the clinical rationale, system architecture, uncertainty quantification methodology, and a staged validation framework encompassing retrospective evaluation on public datasets (MIMIC-CXR, CheXpert, RSNA-PE), a prospective single-site pilot, and a pragmatic multi-site trial. Primary endpoints include median time-to-read for critical cases, calibration error (ECE <= 0.05), alert positive predictive value >= 0.90, and reduction in missed critical findings.

Keywords

uncertainty quantification, radiology AI, medical image triage, Monte Carlo Dropout, calibrated confidence, chest radiography, computed tomography, clinical decision support, Vision Transformer, PACS integration

1. Problem Statement

1.1 Clinical Burden

Radiology departments globally face an expanding volume-complexity mismatch. In the United States alone, diagnostic imaging volume has grown approximately 5% annually over the past decade, while the radiologist workforce has not kept pace [1]. Emergency departments depend on timely identification of critical findings -- pneumothorax, pulmonary embolism, intracranial hemorrhage, stroke -- where delays of even 30-60 minutes can alter patient outcomes [2]. Worklist prioritization remains largely manual or rule-based, creating a system where a critical pneumothorax on a routine outpatient chest X-ray may wait behind non-urgent cases.

1.2 The Binary Output Problem

Current commercial radiology AI systems (e.g., Aidoc, Viz.ai, Qure.ai) have achieved regulatory clearance for specific findings, yet most produce binary outputs: "positive" or "negative," sometimes accompanied by a probability score that is not calibrated and not actionable [3]. Clinicians report that uncalibrated scores erode rather than build trust, because a stated "92% probability" may not correspond to a true 92% positive rate in the deployment population [4].

1.3 The Trust Deficit

Multiple surveys indicate that the primary barrier to adoption is not diagnostic accuracy per se but rather the inability

to gauge when the AI is likely wrong [6]. Without uncertainty communication, clinicians face two poor choices: trust every AI output (risking false negatives) or verify every output (negating efficiency gains). An uncertainty-aware system that communicates "I am not confident in this case" would allow radiologists to allocate attention where it is most needed.

2. Background and Current Industry Practice

Current radiology workflow follows a first-in-first-out (FIFO) or rules-based priority model. Some institutions use AI triage tools that reorder worklists based on binary predictions. While these reduce median time-to-read for flagged cases, they introduce a new failure mode: false negatives that are actively deprioritized, potentially increasing harm compared to FIFO [7]. The critical architectural gap is the absence of an uncertainty estimation layer between inference and routing.

3. Novelty Statement

This work proposes the integration of epistemic uncertainty quantification into the radiology AI triage pipeline as a first-class system component -- not a post-hoc addition -- that directly governs clinical routing decisions. While uncertainty estimation methods (Monte Carlo Dropout, deep ensembles, evidential deep learning) are established in the ML literature [8,9], their translation into production radiology systems with validated routing logic, calibration guarantees, and PACS-native integration remains unrealized.

Level	What is New
Method	Calibrated uncertainty with ECE targets in training objective
System	Three-tier routing driven by uncertainty, not just probability
Workflow	Senior review queue for low-confidence cases + feedback loop
Interop	DICOM SR output encoding uncertainty metadata

Potential Patentability and Prior-Art Differentiation (Non-Legal)

The combination of (a) calibrated uncertainty scoring, (b) three-tier clinical routing logic, and (c) DICOM SR-encoded uncertainty metadata may constitute a novel system-level invention. All patentability claims herein are preliminary and require professional legal evaluation by qualified patent counsel.

4. Proposed System Architecture

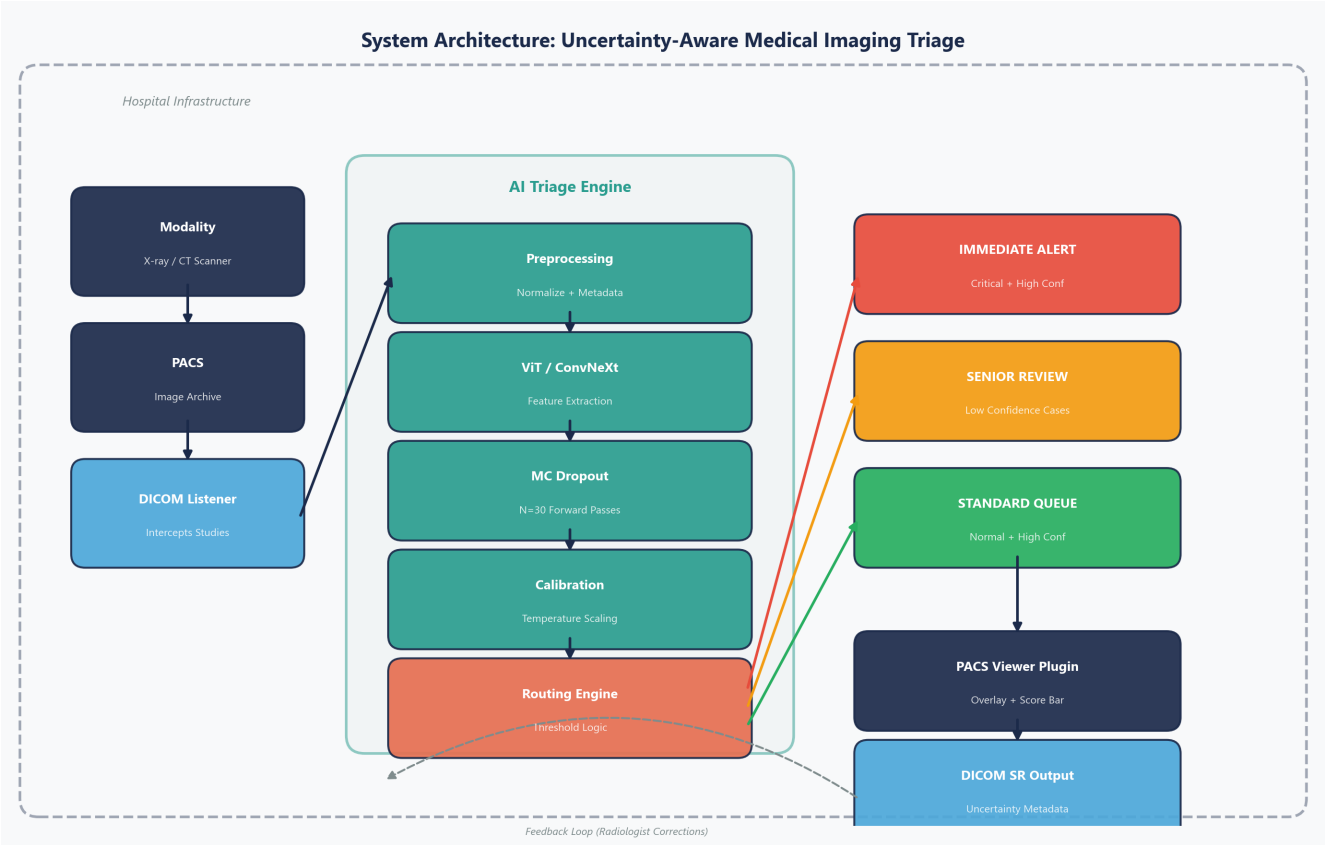

Figure 1: End-to-end system architecture showing DICOM ingestion, AI triage engine with MC Dropout uncertainty estimation, three-tier routing, and PACS viewer integration.

4.1 AI Components

Feature Extraction: Vision Transformer (ViT-B/16 or ViT-L/16) pretrained on ImageNet-21k, fine-tuned on chest radiograph and CT datasets. ConvNeXt-V2 serves as an alternative for latency-constrained settings.

Uncertainty Estimation: Epistemic uncertainty via Monte Carlo Dropout -- at inference, dropout layers remain active and N=30 stochastic forward passes are performed. Predictive mean provides classification; predictive variance provides the uncertainty score [8].

Calibration: Post-hoc temperature scaling on a held-out calibration set to minimize Expected Calibration Error (ECE). Target: ECE <= 0.05 across all pathology classes [12].

4.2 Routing Logic

Prediction	Confidence	Action
Critical finding	High (>= tau)	Immediate alert; top of worklist
Critical finding	Low (< tau)	Senior radiologist review queue
No critical finding	High (>= tau)	Deprioritize (standard queue)
No critical finding	Low (< tau)	Senior radiologist review queue

4.3 Safety Guardrails

- No autonomous action: system reorders worklists and generates alerts only
- Fail-open design: worklist reverts to FIFO if AI engine is unavailable

- Full audit logging of every prediction, score, routing decision, and radiologist action
- Out-of-distribution detection flags atypical studies for manual review

5. Clinical Triage Workflow

Figure 2: Stepwise clinical workflow from study acquisition through AI inference, uncertainty-based routing, and radiologist feedback capture.

6. Uncertainty Quantification Methodology

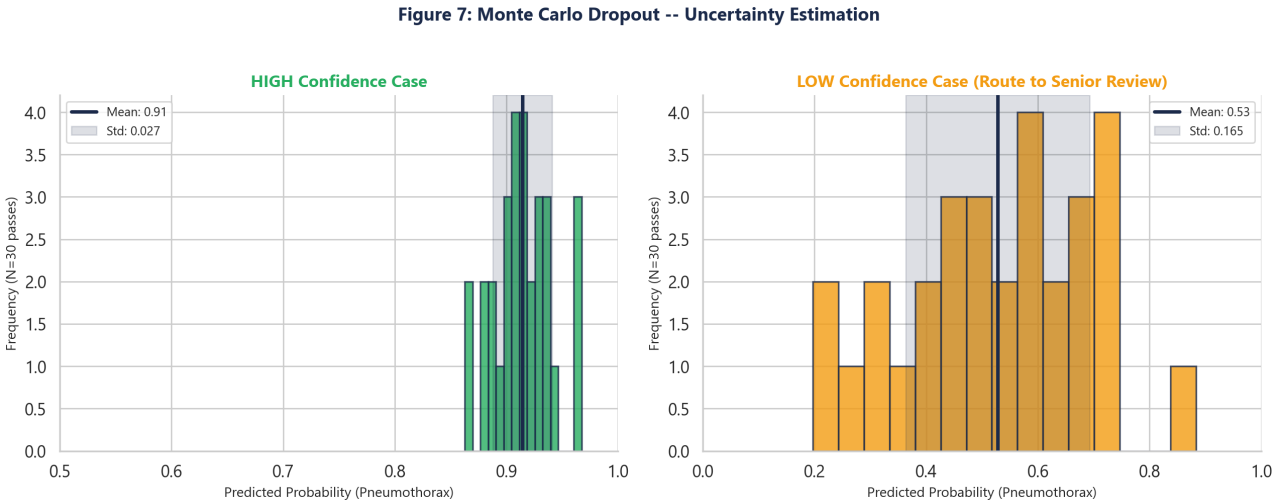

Figure 3: Monte Carlo Dropout illustration. Left: high-confidence case with tight prediction distribution. Right: low-confidence case with wide spread, routed to senior review.

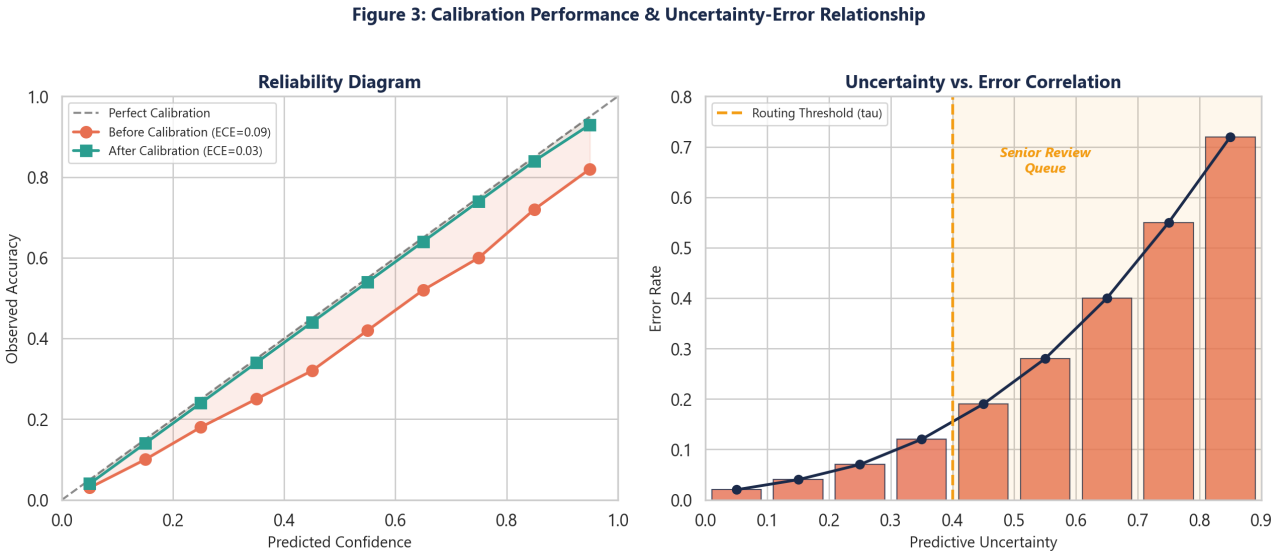

Figure 4: (Left) Reliability diagram showing calibration improvement after temperature scaling. (Right) Correlation between predictive uncertainty and error rate, with routing threshold indicated.

7. Routing Decision Matrix

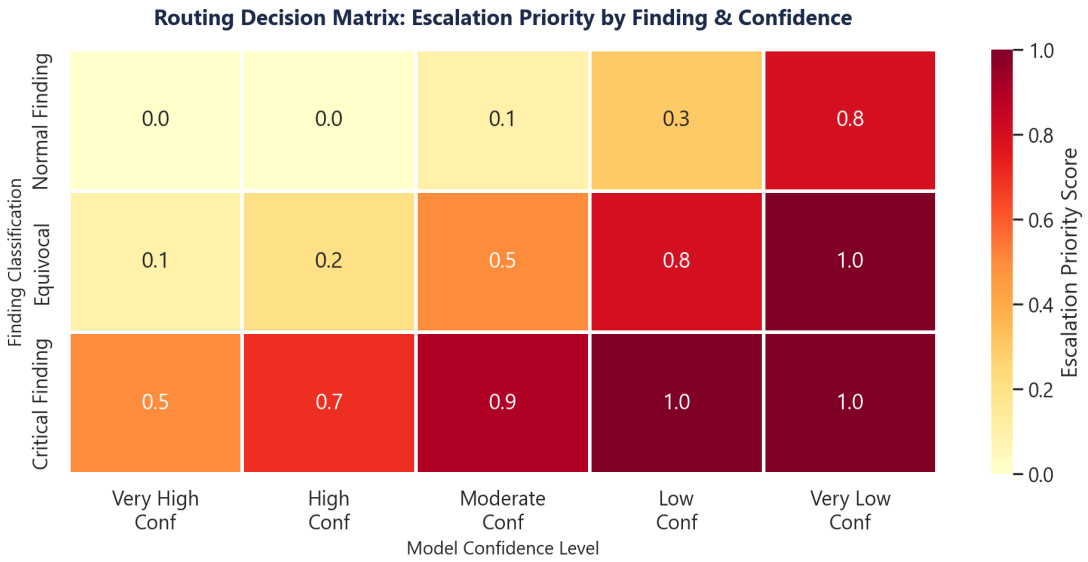

Figure 5: Escalation priority heatmap. Higher scores (darker) indicate greater urgency for radiologist attention. Low-confidence cases are escalated regardless of finding type.

8. Data Sources and Evidence Base

Data Type	Source	Purpose
Chest X-rays	MIMIC-CXR, CheXpert	Training & validation
CT Pulmonary Angiography	RSNA-PE dataset	PE detection model
CT Head	CQ500	Stroke/hemorrhage detection
Radiology reports	Paired with above	Weak supervision labels
Expert annotations	Board-certified radiologists	Ground truth
Clinical outcomes	EHR linkage (prospective)	Endpoint validation

MC Dropout uncertainty estimation has been validated in diabetic retinopathy screening [17], chest X-ray interpretation [18], and histopathology [19]. No prior study has validated a complete uncertainty-driven triage routing system in a prospective clinical setting -- this is the primary evidence gap the proposed validation addresses.

9. Methodology for Validation

Phase 1 -- Retrospective (Months 0-6)

Train and evaluate on public datasets. Primary metrics: AUROC per pathology, ECE, uncertainty-error correlation (Spearman rho >= 0.60), routing simulation.

Phase 2 -- Prospective Silent Pilot (Months 6-12)

Deploy in shadow mode at one academic medical center. AI runs in parallel; predictions logged but not displayed. Compare AI triage ordering with radiologist-determined urgency.

Phase 3 -- Prospective Interventional Pilot (Months 12-18)

Randomized worklist study at 2-3 sites. Intervention: AI-reordered worklist with uncertainty display. Control: standard FIFO. Primary endpoint: median time from acquisition to critical finding communication. Sample size: 1,200 studies per arm (power 0.80, alpha 0.05).

10. Hypothesis-Metric-Validation Framework

Hypothesis	Metric	Data Needed	Validation
Scores calibrated	ECE <= 0.05	Held-out test set	Reliability diagram
Uncertainty ~ errors	Spearman rho >= 0.60	Predictions + GT	Correlation analysis
Routing catches misses	>= 80% FN in low-conf	Retrospective sim	Threshold sweep
Faster critical reads	>= 35% time reduction	Prospective timestamps	Wilcoxon test
Higher trust scores	>= 1.5 Likert increase	Prospective surveys	Paired t-test
Manageable alerts	<= 5 alerts/shift	Alert logs	Descriptive stats

11. Expected Impact

Figure 5: Projected Impact Dashboard

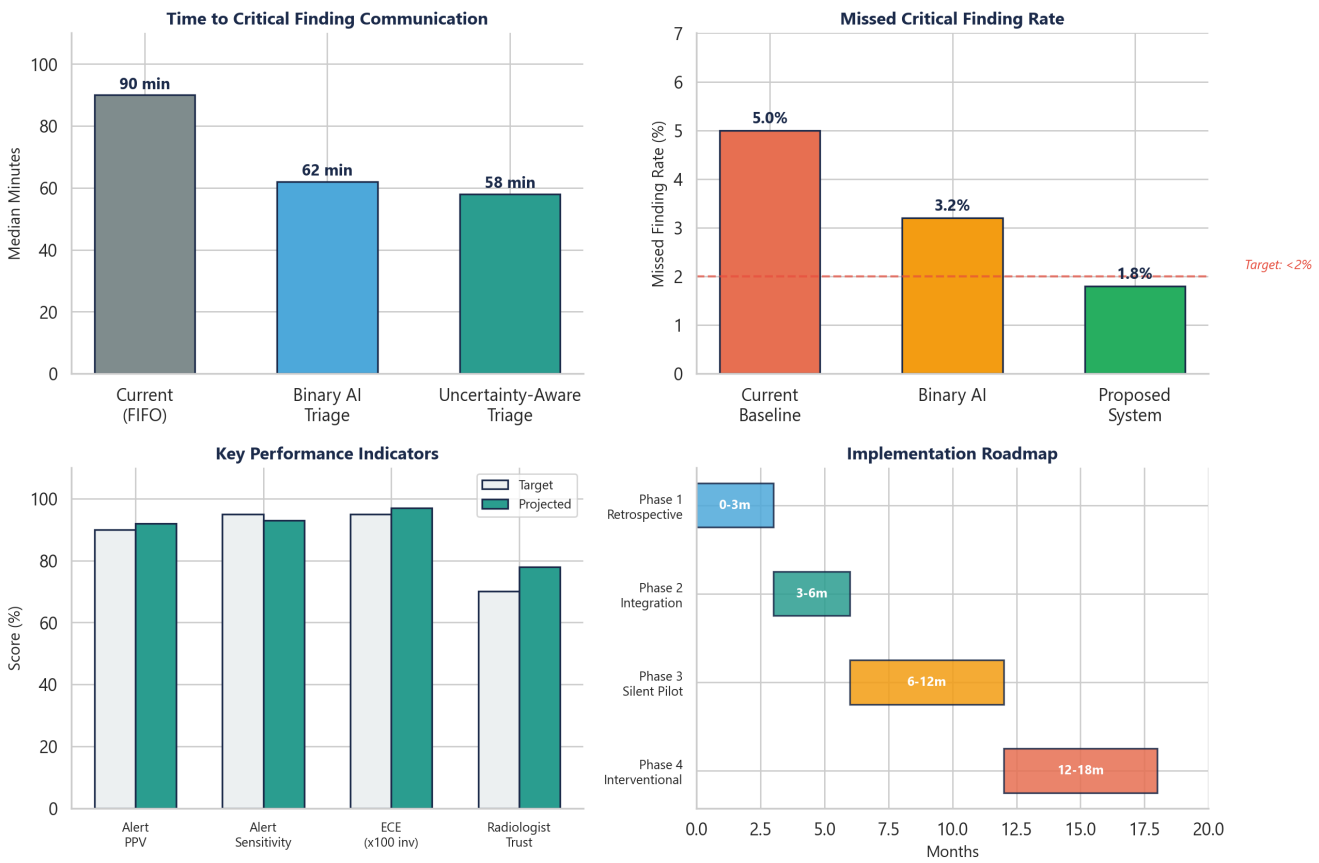

Figure 6: Projected impact dashboard showing (a) time-to-critical-finding reduction, (b) missed finding rate improvement, (c) KPI targets vs projections, (d) implementation timeline.

12. Comparison Against Existing Solutions

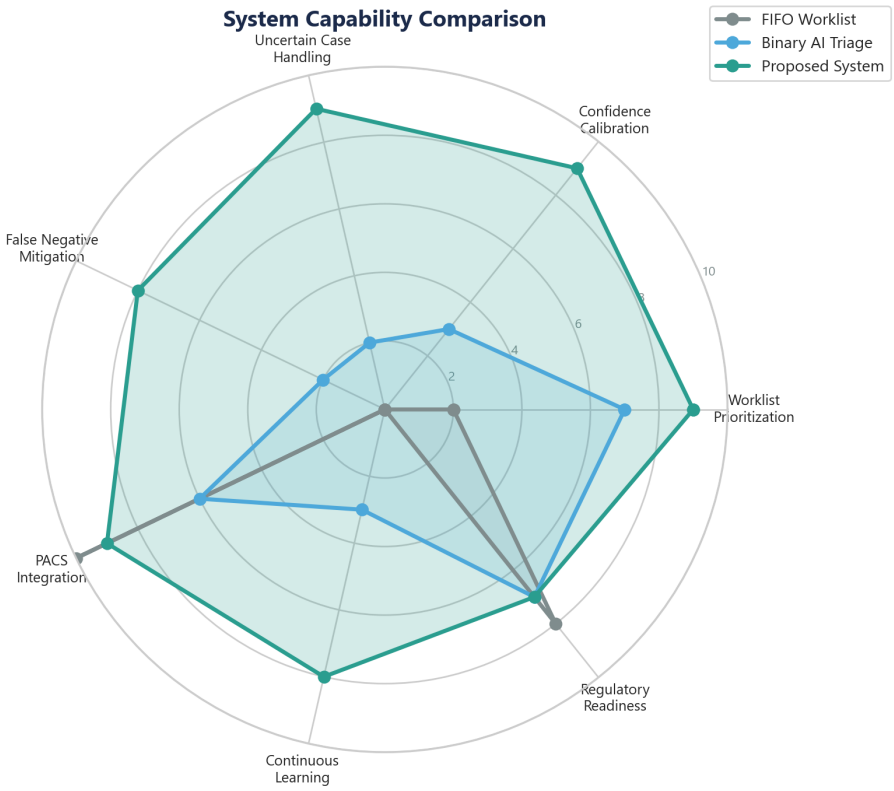

Figure 7: Multi-dimensional capability comparison across FIFO worklist, binary AI triage, and the proposed uncertainty-aware system.

Feature	FIFO	Binary AI	Proposed System
Prioritization	Manual	AI positive->top	AI + uncertainty tiers
Confidence info	None	Uncalibrated	Calibrated (ECE<=0.05)
Uncertain cases	No distinction	Same as confident	Senior review queue
FN mitigation	None	None	Low-conf escalation
PACS integration	Native	Worklist flag	Worklist + overlay + SR
Learning loop	None	Limited	Radiologist feedback

13. Feasibility Analysis

Technical

MC Dropout with N=30 forward passes on ViT-B/16 requires ~3 seconds per study on a single NVIDIA A100 GPU. Stack: Python 3.10+, PyTorch 2.x, pydicom, OHIF Viewer plugin, FastAPI, PostgreSQL.

Clinical Workflow

Integration at worklist level requires no change to interpretation behavior. PACS viewer plugin adds color-coded confidence bar (green/yellow/red) and numerical uncertainty score.

Regulatory & Privacy

FDA 510(k) pathway with predicate comparison to existing cleared triage AI. All processing within hospital network. HIPAA compliance through standard DICOM de-identification. System is advisory only -- positioned as Clinical Decision Support, not autonomous diagnostic software.

14. Implementation Roadmap

Phase	Timeline	Deliverables	Key Risks
Foundation	0-3 months	Baseline model, MC Dropout, calibration pipeline	Minority class calibration
Integration	3-6 months	DICOM listener, API, PACS plugin, audit DB	PACS vendor variability
Silent Pilot	6-12 months	Shadow deployment, drift monitoring, IRB protocol	Distribution shift
Interventional	12-18 months	Multi-site RCT, regulatory dossier	Workflow resistance

15. Limitations and Risks

- Model drift: changes in imaging protocols or demographics may degrade calibration
- Computational overhead: 30 forward passes increases inference time ~30x vs single-pass
- Alert fatigue: conservative thresholds may overload senior review queue
- Bias: calibration quality may vary across demographic subgroups; fairness audits required
- Legal: uncertainty communication may create new liability questions

16. Future Work

- Multi-condition expansion: aortic dissection, bowel obstruction, additional modalities
- Evidential deep learning: deterministic uncertainty to eliminate multi-pass overhead
- Federated learning: multi-site improvement without data sharing
- Adaptive thresholds: context-aware adjustment based on workload and expertise
- LLM integration: couple uncertainty scores with auto-generated preliminary findings

17. Conclusion

The proposed uncertainty-aware medical imaging triage system addresses a fundamental gap in current radiology AI: the absence of honest, calibrated communication about what the model does and does not know. By routing low-confidence predictions to senior radiologists rather than treating all outputs as equally reliable, the system creates a safety net that existing binary-output tools lack. The three-phase validation framework provides a rigorous pathway from concept to clinical evidence. The uncertainty quantification approach described here is not specific to radiology; it represents a design pattern applicable to any clinical AI system where the cost of silent failure exceeds the cost of escalation.

18. References

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. Submission Readiness Checklist

[ ] All co-authors reviewed and approved
[ ] Abstract within npj Digital Medicine word limit (<= 350 words)
[ ] All figures meet journal resolution requirements (300 dpi)
[ ] Architecture and workflow diagrams redrawn as vector graphics
[ ] All "to be validated" placeholders resolved or acknowledged
[ ] Statistical analysis plan reviewed by biostatistician
[ ] IRB approval documentation ready
[ ] Data and code availability statements prepared
[ ] Competing interests declarations collected
[ ] EQUATOR/TRIPOD checklist completed
[ ] Cover letter drafted
[ ] Patent counsel consulted regarding novelty claims